

INFORMAȚII PERSONALE

Traian Florin Șerbănuță

📍 Strada Academiei nr. 14, 010014 București (România)

☎ +40213143508

✉ traian.serbanuta@fmi.unibuc.ro

EXPERIENȚA PROFESIONALĂ

10/2013–Prezent

Conferențiar Universitar

Universitatea din București

01/2011–12/2013

Cercetator Post-doctoral

Universitatea Alexandru Ioan Cuza, Iasi (România)

01/2011–07/2012

Post Doctoral Researcher

University of Illinois

08/2004–12/2010

Research assistant

University of Illinois

05/2007–08/2007

Intern

Google, New York (Statele Unite ale Americii)

05/2005–08/2005

Research intern

Microsoft Research, Redmond (Statele Unite ale Americii)

10/2003–06/2004

Preparator

Universitatea din București

2000–2001

IT Developer

Popnet Agentscape, București (România)

EDUCAȚIE ȘI FORMARE

08/2004–02/2010

PhD in Computer Science

University of Illinois at Urbana-Champaign

06/06/2004–19/06/2004

Summer School Logics of Formal Software Specification Languages, Stara Lesna (Slovakia)

10/2002–02/2004

Master în Informatică

Universitatea din București

10/1998–06/2002

Licențiat în Informatică

Universitatea din București

COMPETENȚE PERSONALE

Limba(i) maternă(e)

română

Alte limbi străine cunoscute

	ÎNȚELEGERE		VORBIRE		SCRIERE
	Ascultare	Citire	Participare la conversație	Discurs oral	
engleză	C2	C2	C2	C2	C2
franceză	B1	B1	A1	A2	A2

Niveluri: A1 și A2: Utilizator elementar - B1 și B2: Utilizator independent - C1 și C2: Utilizator experimentat
 Cadrul european comun de referință pentru limbi străine

ANEXE

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Traian Florin Șerbănuță — Lista lucrărilor științifice

Universitatea din București, Departamentul de Informatică

10 martie 2017

A. Teza de doctorat

- [1] Traian Florin Șerbănuță. *A Rewriting Approach to Concurrent Programming Language Design and Semantics*. PhD thesis, University of Illinois at Urbana-Champaign, December 2010.

B. Cărți și capitole în cărți

- [1] Traian Florin Șerbănuță. *Programming Language Design and Analysis: a rewriting approach*. Editura Universității „Alexandru Ioan Cuza”, 2012. Această carte este o reeditare revizuită a tezei de doctorat a autorului.
- [2] Dorel Lucanu, Traian-Florin Șerbănuță, and Grigore Roșu. Towards a $\mathbb{K}$ ool future. In *Theory and Practice of Formal Methods*, pages 325–343. Springer International Publishing, 2016.

C. Lucrări indexate ISI

- [1] Vlad Rusu, Dorel Lucanu, Traian-Florin Șerbănuță, Andrei Arusoaie, Andrei Ștefănescu, and Grigore Roșu. Language definitions as rewrite theories. *Journal of Logical and Algebraic Methods in Programming*, 85(1):98–120, 2016.
- [2] Traian Florin Șerbănuță and Liviu Dinu. Maximally parallel contextual string rewriting. In *Rewriting Logic and Its Applications - 11th International Workshop, WRLA 2016*, LNCS, 2016.
- [3] Claudia Elena Chiriță and Traian Florin Șerbănuță. An institutional foundation for the K semantic framework. In *Recent Trends in Algebraic Development Techniques (WADT'14)*, volume 9463 of LNCS, pages 9–29. Springer, 2015.

- [4] Philip Daian, Ylies Falcone, Patrick Meredith, Traian Florin Șerbănuță, Akihito Iwai, Shin'ichi Shiriashi, and Grigore Roșu. RV-Android: Efficient parametric Android run-time verification, a brief tutorial. In *Runtime Verification (RV'15)*, volume 9333 of *LNCS*, pages 342–357. Springer, 2015.
- [5] Andrei Arusoaie, Dorel Lucanu, Vlad Rusu, Traian-Florin Serbanuta, Andrei Stefanescu, and Grigore Rosu. Language definitions as rewrite theories. In Santiago Escobar, editor, *Rewriting Logic and Its Applications*, Lecture Notes in Computer Science, pages 97–112. Springer International Publishing, 2014.
- [6] Qingzhou Luo, Yi Zhang, Choonghwan Lee, Dongyun Jin, Patrick O'Neil Meredith, Traian Florin Șerbănuță, and Grigore Roșu. RV-Monitor: Efficient parametric runtime verification with simultaneous properties. In *Runtime Verification (RV'14)*, volume 8734 of *LNCS*, pages 285–300. Springer, 2014.
- [7] Regina Frei, Traian Florin Șerbănuță, and Giovanna Di Marzo Serugendo. Self-organising assembly systems formally specified in Maude. *Journal of Ambient Intelligence and Humanized Computing*, 5:491–510, 2014.
- [8] Regina Frei, Giovanna Di Marzo Serugendo, and Traian Florin Șerbănuță. Ambient intelligence in self-organising assembly systems using the chemical reaction model. *Journal of Ambient Intelligence and Humanized Computing*, 1(3):163–184, 2010.
- [9] Grigore Roșu and Traian Florin Șerbănuță. An overview of the K semantic framework. *Journal of Logic and Algebraic Programming*, 79(6):397–434, 2010.
- [10] Traian Florin Șerbănuță and Grigore Roșu. K-Maude: A rewriting based tool for semantics of programming languages. In Peter Csaba Ölveczky, editor, *Rewriting Logic and Its Applications - 8th International Workshop, WRLA 2010*, volume 6381 of *Lecture Notes in Computer Science*, pages 104–122, 2010.
- [11] Camelia Chira, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. P systems with control nuclei: The concept. *Journal of Logic and Algebraic Programming*, 79(6):326–333, 2010.
- [12] Traian Florin Șerbănuță, Grigore Roșu, and José Meseguer. A rewriting logic approach to operational semantics. *Information and Computation*, 207:305–340, 2009.
- [13] Chucky Ellison, Traian Florin Șerbănuță, and Grigore Roșu. A rewriting logic approach to type inference. In *Recent Trends in Algebraic Development Techniques — 19th International Workshop, WADT 2008, Pisa, Italy, June 13-16, 2008, Revised Selected Papers*, volume 5486 of *Lecture Notes in Computer Science*, pages 135–151. Springer, 2009.

- [14] Andrei Popescu, Traian Florin Șerbănuță, and Grigore Roșu. A semantic approach to interpolation. *Journal of Theoretical Computer Science*, 410:1109–1128, 2009.
- [15] Traian Florin Șerbănuță, Gheorghe Ștefănescu, and Grigore Roșu. Defining and executing P systems with structured data in K. In David W. Corne, Pierluigi Frisco, Gheorghe Paun, Grzegorz Rozenberg, and Arto Salomaa, editors, *Workshop on Membrane Computing (WMC'08)*, volume 5391 of *Lecture Notes in Computer Science*, pages 374–393. Springer, 2009.
- [16] Grigore Rosu, Wolfram Schulte, and Traian Florin Șerbănuță. Runtime verification of C memory safety. In Saddek Bensalem and Doron A. Peled, editors, *Runtime Verification (RV'09)*, volume 5779 of *Lecture Notes in Computer Science*, pages 132–152, 2009.
- [17] Feng Chen, Traian Florin Șerbănuță, and Grigore Roșu. jPredictor: a predictive runtime analysis tool for Java. In *ICSE '08: Proceedings of the 30th International Conference on Software Engineering*, pages 221–230, New York, NY, USA, 2008. ACM.
- [18] Virgil Nicolae Șerbănuță and Traian Florin Șerbănuță. Injectivity of the Parikh matrix mappings revisited. *Fundamenta Informaticae*, 73:265–283, 2006.
- [19] Andrei Popescu, Traian Șerbănuță, and Grigore Rosu. A semantic approach to interpolation. In *Proceedings of the 9th International Conference on Foundations of Software Science and Computation Structures, FOSSACS 2006, Vienna, Austria, March 25-31, 2006*, volume 3921 of *Lecture Notes in Computer Science*, pages 307–321. Springer, 2006. also appeared as Technical Report UIUCDCS-R-2005-2643, May 2005.
- [20] Traian Florin Șerbănuță and Grigore Roșu. Computationally equivalent elimination of conditions - extended abstract. In *Proceedings of Rewriting Techniques and Applications (RTA'06)*, volume 4098 of *Lecture Notes in Computer Science*, pages 19–34. Springer, 2006. also appeared as Technical Report UIUCDCS-R-2006-2693, February 2006.
- [21] Traian Florin Șerbănuță. Extending Parikh matrices. *Journal of Theoretical Computer Science*, 310(1-3):233–246, 2004.

D. Lucrări publicate în reviste și volume de conferințe cu referenți (neindexate)

- [1] Philip Daian, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Șerbănuță, and Grigore Roșu. Runtime verification at work: A tutorial. In *International Conference on Runtime Verification*, pages 46–67. Springer International Publishing, 2016.

- [2] Andrei Ștefănescu, Ștefan Ciobăcă, Radu Mereuta, Brandon M Moore, Traian Florin Șerbănuță, and Grigore Roșu. All-path reachability logic. In *Rewriting and Typed Lambda Calculi (RTA'14)*, volume 8560 of *Lecture Notes in Computer Science*, pages 425–440. Springer, 2014.
- [3] Grigore Roșu and Traian Florin Șerbănuță. K overview and simple case study. In Mark Hills, editor, *Proceedings of K'11*, volume 304 of *Electronic Notes in Theoretical Computer Science*, pages 3–56. Elsevier, 2014.
- [4] Traian Florin Șerbănuță. Rewriting semantics and analysis of concurrency features for a C-like language. In Mark Hills, editor, *Proceedings of K'11*, volume 304 of *Electronic Notes in Theoretical Computer Science*, pages 167–182, 2014.
- [5] Traian Florin Șerbănuță, Andrei Arusoaie, David Lazar, Chucky Ellison, Dorel Lucanu, and Grigore Roșu. The K primer (version 3.3). In Mark Hills, editor, *Proceedings of K'11*, volume 304 of *Electronic Notes in Theoretical Computer Science*, pages 57–80, 2014.
- [6] Traian Florin Șerbănuță, Feng Chen, and Grigore Roșu. Maximal causal models for sequentially consistent systems. In Shaz Qadeer, editor, *Runtime Verification (RV'12)*, volume 7687 of *Lecture Notes in Computer Science*, pages 136–150, 2013.
- [7] Traian Florin Șerbănuță and Grigore Roșu. A truly concurrent semantics for the K framework based on graph transformations. In Hartmut Ehrig, Gregor Engels, Hans-Jörg Kreowski, and Grzegorz Rozenberg, editors, *ICGT*, volume 7562 of *Lecture Notes in Computer Science*, pages 294–310. Springer, 2012.
- [8] David Lazar, Andrei Arusoaie, Traian Florin Șerbănuță, Chucky Ellison, Radu Mereuta, Dorel Lucanu, and Grigore Rosu. Executing formal semantics with the K Tool. In Dimitra Giannakopoulou and Dominique Méry, editors, *FM*, volume 7436 of *Lecture Notes in Computer Science*, pages 267–271. Springer, 2012.
- [9] Dorel Lucanu, Traian Florin Șerbănuță, and Grigore Roșu. K Framework distilled. In *9th International Workshop on Rewriting Logic and its Applications, WRLA, 2012*, volume 7571 of *Lecture Notes in Computer Science*, pages 31–53. Springer, 2012. Invited talk.
- [10] Andrei Arusoaie, Traian Florin Șerbănuță, Chucky Ellison, and Grigore Roșu. Making Maude definitions more interactive. In *9th International Workshop on Rewriting Logic and its Applications, WRLA 2012*, volume 7571 of *Lecture Notes in Computer Science*, pages 83–98. Springer, 2012.
- [11] Sudipto Ghoshal, Solaiappan Manimaran, Grigore Roșu, Traian Florin Șerbănuță, and Gheorghe Ștefănescu. Monitoring IVHM systems using a monitor-oriented programming framework. In *The Sixth NASA Langley Formal Methods Workshop (LFM 2008)*, 2008.

- [12] Traian Florin Șerbănuță, Grigore Rosu, and José Meseguer. A rewriting logic approach to operational semantics (extended abstract). In *Structural Operational Semantics 2007 (SOS'07)*, volume 192 of *ENTCS*, pages 125–141. Elsevier, 2007. also appeared as Technical Report UIUCDCS-R-2006-2780, October 2006.
- [13] Mark Hills, Traian Florin Șerbănuță, and Grigore Roșu. A rewrite framework for language definitions and for generation of efficient interpreters. In *Proceedings of the 6th International Workshop on Rewriting Logic and its Applications (WRLA'06)*, volume 176 of *Electronic Notes in Theoretical Computer Science*, pages 215–231. Elsevier Science, July 2007. also appeared as Technical Report UIUCDCS-R-2005-2667, December 2005.
- [14] David Reitter, Ștefan Covaci, Florin Oltean, Cătălin Băcanu, and Traian Florin Șerbănuță. Hybrid natural language processing in a customer-care environment. In *Proceedings of the 11th Student Conference on Computational Linguistics (TaCoS'01)*, 2001.